indicate that the lead incorporated in the black shale at the time of its deposition was normal in its isotopic composition. His hypothesis that uranium has been continuously introduced into the kolm by percolating ground water is not supported by the observed distribution of uranium in the kolm. Further, it should be pointed out that the effect on the isotopic ages due to introduction of uranium is indistinguishable from that due to loss of lead. Therefore, even if this hypothesis were adopted it would not cause any revision in the interpretation that the minimum age of the kolm is about 500 million years. Davidson's deduction that the formation is necessarily younger than 650 million years since this is the lowest lead-207–lead-206 age is not valid. The meaning of this age is directly dependent on the nature of the chemical alteration that occurred.

The current investigations in a number of laboratories on the age of stratigraphically well-defined rocks will permit a much closer definition of the geological time-scale in the near future. The extended scale proposed by Mayne *et al.*[1] is based on limited data and is not supported by measurements other than their own.

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[1] Mayne, K. I., Lambert, R. St. J., and York, D., *Nature*, **183**, 212 (1959).
[2] Holmes, A., *Trans. Geol. Soc. Glasgow*, **21**, 117 (1947).
[3] Davidson, C. F., *Nature*, **183**, 768 (1959). Mayne, K. I., Lambert, R. St. J., and York, D., *Nature*, **183**, 769 (1959).
[4] Miller, D. S., and Kulp, J. L., *Trans. Amer. Geophys. Union*, **39**, 525 (1958).
[5] Miller, D. S., and Kulp, J. L., *Econ. Geol.*, **53**, 937 (1958).
[6] Kulp, J. L., and Long, L. E., *Trans. Amer. Geophys. Union*, **39**, 522 (1958).
[7] Long, L. E., Kulp, J. L., and Eckelmann, F. D., *Amer. J. Sci.* (in the press).
[8] Kruger, Fredrick C., *Geol. Soc. Amer. Bull.*, **57**, 161 (1946).
[9] Damon, P. E., and Kulp, J. L., *Amer. J. Sci.*, **255**, 697 (1957).
[10] Hurley, P., *et al.*, Atom. Energy Comm. Rep. *NYO* 3938 (December 1958).
[11] Adams, J. A. S., Edwards, G., Henle, W., and Osmond, K., *Geol. Soc. Amer. Bull.*, **69**, 1527 (1958).
[12] Cobb, J. C., and Kulp, J. L., *Geol. Soc. Amer. Bull.*, **69**, 1547 (1958).
[13] Cobb, J. C., and Kulp, J. L. (to be submitted to *Geochim. et Cosmochim. Acta*).

Palæomagnetism and Rotation of Newfoundland

Nairn, Frost and Light have recently reported[1] on the palæomagnetic results from Carboniferous and Pre-Cambrian rocks of Newfoundland. By comparing their Carboniferous results with results obtained by other workers[2] in the western United States they show that Newfoundland may have been rotated 20° anticlockwise since Carboniferous time.

I wish to report some results obtained from Carboniferous rocks of New Brunswick and the Gaspé Peninsula, which, with the results of Nairn *et al.*, show that no relative rotation between Newfoundland and these areas has occurred since Carboniferous times. Therefore, if the proposed rotation of Nairn *et al.* is correct, New Brunswick, the Gaspé Peninsula and Newfoundland must have rotated as a unit.

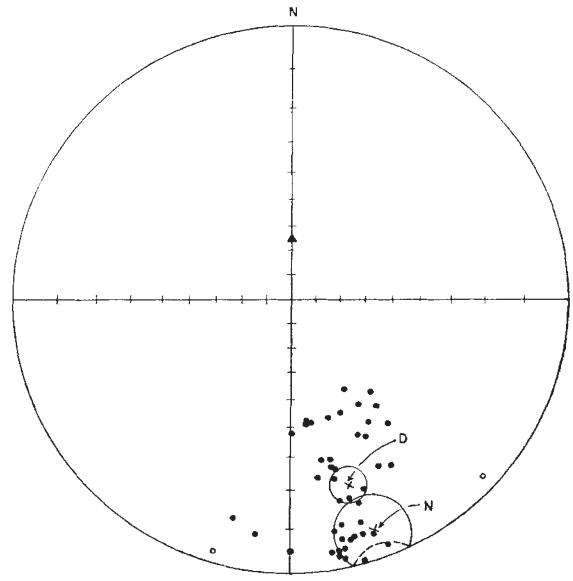


Fig. 1. Directions of magnetization measured in Carboniferous rocks of the Canadian Maritime Provinces. ●, Directions of magnetization of my samples, north pole on lower hemisphere; ○, directions of magnetization of my samples, north pole on upper hemisphere; $X \leftarrow D$, mean direction of my samples (with circle of confidence at Fisher's 95 per cent level of probability); $X \leftarrow N$, mean direction of samples of Nairn *et al.* (with circle of confidence at Fisher's 95 per cent level of probability); ▲, direction of magnetic field produced by a central axial dipole oriented along the Earth's present axis of rotation

In the summer of 1958 twenty-two samples of the Bonaventure formation (Upper Mississippian or Lower Pennsylvanian) were collected by me from the south shore of the Gaspé Peninsula, Province of Quebec, between Percé and Maguasha. Fourteen samples were also collected from the Kennebecasis formation of Mississippian age at localities about fifteen miles north-east of St. John, New Brunswick. Ten samples were collected by Dr. C. H. Stockwell in the autumn of 1955 from the Pennsylvanian Bathurst formation near Bathurst, New Brunswick. Two 1-in. cubes were cut from each of these samples and were measured on the astatic magnetometer of the Dominion Observatory, Ottawa. The mean direction of each sample has been plotted on a stereogram in Fig. 1. Since rocks were collected from an extensive area, their directions have been corrected to the geographical location 48° N. and 66° W. by assuming that the Earth's average magnetic field is a dipole. The mean direction of the results of Nairn *et al.* from the Codroy gypsum also has been corrected to 48° N. and 66° W. and plotted on Fig. 1.

By inspection one can see that there is no significant difference between the declinations of the Newfoundland samples and of those reported here. This surmise is confirmed by Fisher's statistical analysis which gives a mean direction, corrected to 48° N. and 66° W., of $D = 160{\cdot}5°$ and $I = +6{\cdot}0°$ with a circle of confidence of 8·4° for the samples of Nairn *et al.*, and a mean direction of $D = 163{\cdot}5°$ and $I = +19{\cdot}5°$ with a circle of confidence of 5·0° for my samples. Thus, there is no significant difference between the two directions, and there is no evidence for any relative rotation between Newfoundland and the mainland Maritime Provinces, since Carboniferous time.

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[1] Nairn, A. E. M., Frost, D. V., and Light, B. G., *Nature*, **183**, 596 (1959).
[2] Creer, K. M., Irving, E., and Runcorn, S. K., *Proc. Roy. Soc.*, A, **250**, 144 (1957).

BIOCHEMISTRY

Kinetic Study of Dextransucrase based on the Langmuir Adsorption Isotherm

The action of the polysaccharide-synthesizing enzyme dextransucrase has been described previously in terms of *p*H [1] and the effect of various alternate receptors[2,3]. We have now examined the dependence of rate of reaction on the concentration of enzyme. The results are interpreted on the basis of adsorption kinetics[4], where the substrate (sucrose) and acceptor are adsorbed simultaneously on adjacent sites of the enzyme. This is followed by glucosyl transfer from sucrose to acceptor. No reaction occurs between bound sucrose and acceptor in solution, and there is no separate hydrolytic step prior to the transfer of the glucosyl group.

The kinetic studies were carried out by exposing sucrose solutions to varying amounts of crude enzyme solution at 25° C. and *p*H 5·2, and measuring the amount of fructose produced at a given time. The enzyme was prepared as reported elsewhere[1]. Sucrose solution (0·2 ml. of a 0·01 *M* solution in 0·05 *M* acetate buffer *p*H 5·2) was incubated with 0·02–1·6 ml. of the enzyme solution (assaying 49 units/ml.) and buffered to make a final volume of 1·8 ml. After ½ hr. the reaction was quenched with 0·1 *M* sodium hydroxide and the reducing sugars determined colorimetrically[5,6]. A plot of rate against enzyme concentration is shown in Fig. 1.

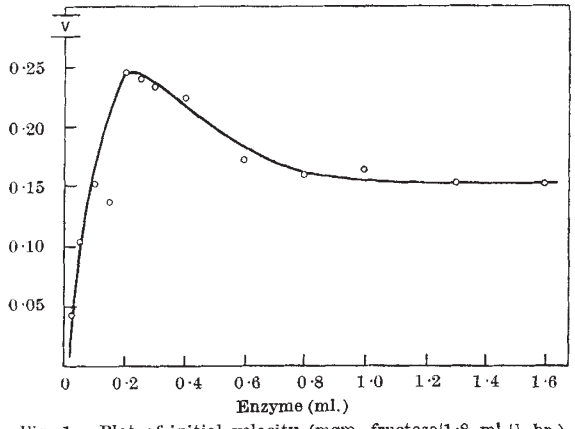


Fig. 1. Plot of initial velocity (mgm. fructose/1·8 ml./½ hr.) against enzyme concentration

The appearance of a maximum in such a plot is rare; however, it has been reported for the lactic dehydrogenase system[7]. Laidler and Socquet[4] have treated this case theoretically and have derived the following rate equation:

where k is the rate constant for the decomposition of the complex; S_1 and S_2 are the two substrates, in this case sucrose and acceptor; K_1 and K_2 are the corresponding equilibrium constants for the formation of the binary complex; K_1' and K_2' are the equilibrium constants for the cross adsorption of S_2 on site 1 and S_1 on site 2 respectively. It is seen that equation (1) predicts linearity of rate against enzyme or sucrose at low concentration, but an inverse relationship and hence a maximum at high concentration.

At high enzyme concentration, the substrates are spread out over a large number of enzyme sites. The observed decrease in rate is attributed to the decreased probability that adjacent sites will be occupied in these circumstances. The previously observed maximum in the rate against sucrose concentration[8] indicates that sucrose is cross-adsorbed on the acceptor sites (which may well be identical with the donor sites) and thus inhibits the reaction. Attempts to observe maxima in S_2 by adding alternate acceptor have been complicated by the presence of the natural acceptor. Thus far, only a roughly linear increase of rate with acceptor concentration has been observed.

If a reaction were occurring between bound substrate and unbound acceptor, no maximum would occur. Furthermore, it has been shown[4] that in cases where one substrate is solvent there is no maximum in the enzyme curve. Hence the possibility of a hydrolytic reaction with sucrose followed by a rate-determining combination with dextran is eliminated. The presence of the observed maxima thus fits only the case of simultaneous two-substrate adsorption.

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[1] Neely, W. B., *J. Amer. Chem. Soc.*, **80**, 2010 (1958).
[2] Koepsell, H. J., Tsuchiya, H. M., Hellman, N. M., Kazenko, A., Hoffman, C. A., Sharpe, E. S., and Jackson, R. W., *J. Biol. Chem.*, **200**, 793 (1952).
[3] Bailey, R. W., Barker, S. A., Bourne, E. J., Grant, P. M., and Stacey, M., *J. Chem. Soc.*, 1895 (1958).
[4] Laidler, K. J., and Socquet, Irene M., *J. Phys. and Coll. Chem.*, **54**, 530 (1950).
[5] Somogyi, M., *J. Biol. Chem.*, **160**, 61 (1945).
[6] Nelson, N., *J. Biol. Chem.*, **153**, 375 (1944).
[7] Socquet, Irene M., and Laidler, K. J., *Arch. Biochem. Biophys.*, **25**, 171 (1950).
[8] Neely, W. B., *Nature*, **182**, 1007 (1958).

Agar Electrophoresis of Normal Soluble Proteins in Guinea Pig Liver

Hitherto, neither free electrophoresis[1] nor electrophoresis on filter paper of the soluble proteins in liver (ref. 2, and Smetana, R., and Kořínek, J., personal communication) has afforded satisfactory delimitation of a sufficient number of fractions. After obtaining a number of unsatisfactory results on filter paper, we turned our attention to electrophoresis on agar gel.

Healthy guinea pigs weighing between 300 and 900 gm. were lightly anæsthetized with ether before being perfused with normal saline. The liver was ground in a mortar with quartz sand. 1–3 ml. of a 0·7 per cent solution of sodium chloride (the amount depending on the size of the liver) was added and the whole stored in a refrigerator. 24 hr. later it was centrifuged at 6,000 r.m.p. for 2–3 hr.

$$v = \frac{kK_1K_2[E]_0[S_1]_0[S_2]_0}{\{1+[E]_0(K_1+K_1')+K_1[S_1]_0+K_1'[S_2]_0\}\{1+[E]_0(K_2+K_2')+K_2'[S_1]_0+K_2[S_2]_0\}} \quad (1)$$

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